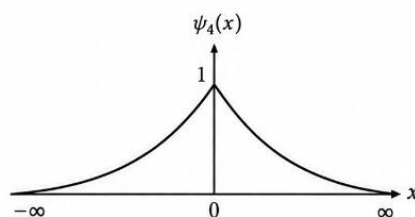
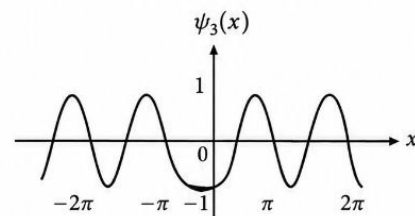
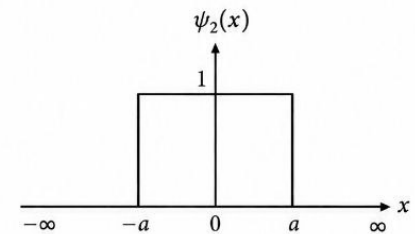
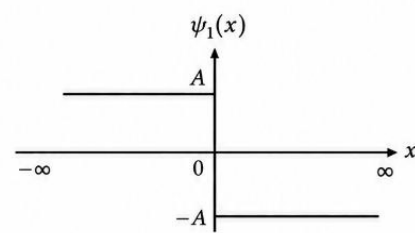
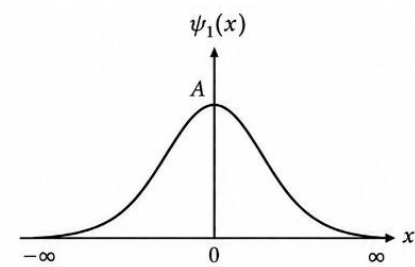


QUIZ 1 (18/8/2026)

Instructions: Do your work neatly and write only in the space provided. TAs and instructor are free NOT to evaluate messy work or answers written out of their place. First, write the solution in your book and only when you are confident to put it on the question.

Maximum Marks [10 points]

1. For the possible functions shown below, which is a valid wavefunction and provide its justification to the right of the figure. ($A = 1$) **[1 point each]**



2. In many semiconductors, the E - k relationship significantly deviates from $E = \frac{\hbar^2 k^2}{2m^*}$. In such cases, it is approximated by

$$\frac{\hbar^2 k^2}{2m^*} = E(1 + \alpha E) \quad (1)$$

What will be the impact of the modified E - k relationship (1) on the density of states for a 2D material ?

[5 Points]